Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 9/22/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

   Yes. We met with Dr. Rabia Shahid on Tuesday, September 23, at 8 pm. Brehon Parker and Anika Baro Villa were present. No other formal team meeting was held this week; communication and updates were maintained through Discord.

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

   Yes. We met with Dr. Ali Ibrahim on Thursday, September 25, at 6 pm. Joshua Adegboyo, Brehon Parker, and Anika Baro Villa were present.

3. Describe verbally the tasks completed the past week and the challenges faced.

   This week (Week 6) we applied unsupervised clustering methods as another way to explore feature selection and reveal structure in the miRNA dataset. Each team member implemented a different algorithm: Anika worked with K-Means, Brehon used Spectral Clustering, Joshua tried Gaussian Mixture Models, and Nicholas applied Hierarchical Clustering. We followed a shared workflow so the results could be compared fairly. For every method we first transposed the data so that each row represented a single miRNA across all patient samples and then standardized those values to ensure that every patient column contributed equally to the distance calculations. We then ran each algorithm with two clusters to separate the miRNAs into groups of potential biological relevance and evaluated the quality of these groupings using metrics such as silhouette scores, while visualizing the results with two-dimensional PCA plots and saving the cluster assignments to CSV files. Finally, each member examined the two resulting miRNA clusters from their own method and selected the one judged to be biologically important.

4. Describe the tasks to be completed the coming week.

   Next week (Week 7) our team will compare the important clusters identified by the unsupervised methods and, where appropriate, explore a second stage cascade clustering to look for finer substructure. In parallel, we will continue and refine the deep learning feature selection work by extending the models we each developed earlier. Each member will keep improving the architecture they previously worked on, whether a feed-forward network, an LSTM, a CNN, a transformer, or a variation of the autoencoder, to capture stronger patterns in the miRNA data and highlight the most informative features for breast cancer stage classification. By comparing these refined deep learning approaches with the classical and unsupervised methods we have already used, we aim to build a more reliable set of key miRNA biomarkers.

Additionally, we will prepare the slides for our first project demo, which will summarize our project's goals, the progress made so far, and the key results from all stages of our analysis. The purpose is to communicate how the combined feature selection work we have done so far supports the development of a breast cancer stage classification system and to outline the next steps toward completing the project.

Previous Report Summary. Date: 9/15/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	2	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	1	2

Current Report Summary. Date: 9/22/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2		
Anika Baro Villa	3	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	0	2

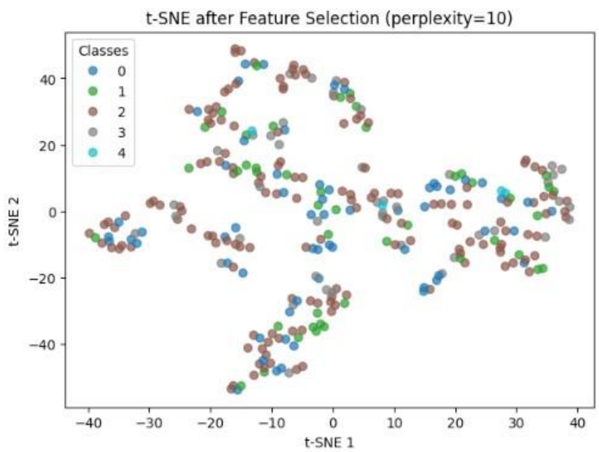
Name of the Member: **Joshua Adegboyo**

Report Date: **9/22/25**

Task 5.1: Apply deep learning for feature selection (9/20/25)

I worked with the same breast cancer miRNA dataset (1,209 samples, 1,881 features, 5 classes). After standardization, I applied **L**assoCV and Chi² feature selection, reducing the feature space to the top 50 miRNAs. I visualized the reduced space using t-SNE, which showed partial but imperfect stage separation.

For classification, I trained an LSTM-based sequential model in Keras, treating the 50 miRNAs as ordered features. The model included stacked LSTM layers with dropout and a softmax output. While LSTM did not outperform simpler models, this experiment highlights both the potential and the limitations of applying sequential deep learning to non-time-series omics data.



Task 5.2: Review of Lasso-Selected and LSTM-Selected miRNAs

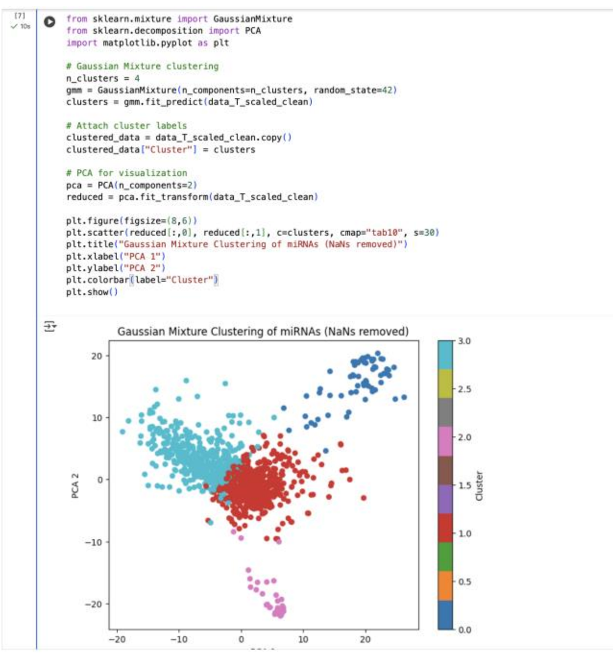
For this task I applied **LassoCV feature selection** in combination with **Chi² filtering** to reduce the highdimensional miRNA dataset to a smaller subset of informative features. These reduced feature sets (e.g., top 50 miRNAs) were then modeled using an **LSTM-based sequential neural network**, which is well suited to capture dependencies among expression profiles. I also generated **t-SNE visualizations** to assess how well the selected features separate the different cancer stages.

From the Lasso/LSTM runs, I identified several candidate miRNAs that consistently appeared across reduced feature sets and contributed to classification, despite the challenge of class imbalance. I reviewed these candidates against the existing cancer biology literature and found that multiple miRNAs highlighted by the model already have known associations with tumor progression, signaling pathways, and therapy response, while others remain less studied. This provides both a validation of our modeling pipeline and a set of exploratory biomarkers for further analysis.

Task 6.1: Apply Gaussian Mixture Model (GMM) clustering for feature selection (9/27/25)

For this task I applied the Gaussian Mixture clustering algorithm to our dataset in order to separate miRNAs into two groups, as instructed by Dr. Ali Ibrahim. I began by transposing the dataset so that each row corresponded to a single miRNA across all patient samples. This ensured that clustering would operate on features rather than patient profiles. I then standardized the data to place all miRNAs on the same scale, preventing those with higher absolute expression ranges from dominating the clustering process.

Using scikit-learn's GaussianMixture implementation, I set the number of components (n_clusters) to 2, following the requirement to split features into "important" vs. "not important" groups. After fitting the model, the algorithm assigned 1880 miRNAs to Cluster 0 and just a single miRNA to Cluster 1. While the distribution was highly imbalanced, this outcome is not unexpected given that many miRNAs in this dataset show similar variability across patients. The important result for now is that the two cluster lists were generated successfully and saved as separate .csv files for comparison with the outputs from the other methods. The consensus of overlapping miRNAs across methods will later determine the final "important" set that Dr. Ali asked us to highlight.



Task 6.2: Determine the important miRNA Cluster through data-driven analysis and literature review (Pending)

In my case, GMM clustering produced one overwhelmingly large cluster and one extremely small cluster (1 miRNA). At this stage, I have not yet carried out further analysis to validate which cluster should be designated as the "important" group. The next step will be to compare the variability of each cluster and to perform a targeted literature search on the single isolated miRNA in Cluster 1. This will allow me to assess whether the model's smaller cluster corresponds to biologically relevant features, or whether additional interpretation is needed.

Task 6.3: Cascade clustering applied to the important cluster (Pending)

Once the important cluster is identified, I will apply cascade clustering to investigate whether it can be subdivided into meaningful subgroups. In my case, this will likely involve running GMM or K-Means again, but restricted to the selected miRNAs, to see whether finer structure emerges. If the silhouette scores remain low or subclusters are indistinct, that will indicate that the cluster is relatively homogeneous and cannot be meaningfully partitioned further.

Task 7.1: Continue Neural Network feature selection tests (10/05/25)

In the next round of assignments I will contribute by helping extend the deep learning based feature selection using feedforward neural networks. My focus will be on testing small changes in architecture and tuning hyperparameters such as layer size and activation functions, and evaluating whether the selected miRNAs overlap with those found by unsupervised clustering methods like GMM.

Task 7.2: Prepare project first demo slides (10/05/25)

I will also assist with preparing our demo slides. This will include documenting the results of my Gaussian clustering work, showing how the miRNAs were partitioned, and summarizing our team's progress across classical, clustering, and neural network based feature selection approaches. The goal will be to provide a clear snapshot of both the workflow completed so far and the path forward for the project.

Name of the Member: Anika Baro Villa

Report Date: 9/22/25

Task 5.1: Apply deep learning feature selection with Feedforward Neural Network (FNN) (9/20/25)

I applied Sequential Feature Selection (SFS) and a simple feed-forward neural network (FNN) to the preprocessed miRNA dataset using the same class-balance checks and macro-F1 scoring as before to make results directly comparable. The FNN, built in Keras with dense layers, dropout, ReLU activations, and a softmax output for five classes (normal tissue plus stages I–IV), was trained with a validation split, early stopping, and a fixed random seed. After training, I ranked miRNAs by the mean absolute value of their input-layer weights and saved the full ranking to fnn_ranked_features.csv. From this ranking I extracted top-k feature sets (k = 10, 8, 5, 3), saved each set to CSV, evaluated them with a downstream multinomial logistic-regression model using 3-fold cross-validation and macro-F1 scoring, and created t-SNE plots to visualize how well the selected features separate the cancer stages. Finally, I produced an overlap summary showing three miRNAs that appeared in every top-k list and five that appeared in at least two lists, giving a concise consensus of consistently important features.

Task 5.2: Perform literature review of stable miRNAs selected by SFS and FNN (9/20/25)

I reviewed the 16 miRNAs that were repeatedly selected (appearing in at least two of the k-runs from both the SFS and the FNN analyses) to evaluate their scientific support as reliable features for our classification model. Using peer-reviewed sources, I examined each miRNA's known cellular functions, reported links to breast cancer, and any potential clinical applications. Of the 16, nine have strong experimental evidence and well-established roles in key cancer pathways including PI3K/AKT/ERK signaling, cell-cycle control, apoptosis, EMT/metastasis, and therapy response; three have only preliminary or limited data from early prognostic or drug-resistance studies; and the remaining four are largely unexplored, appearing so far only in sequencing datasets or predictive databases.

Task 6.1: Apply K-Means clustering for feature selection (9/25/25)

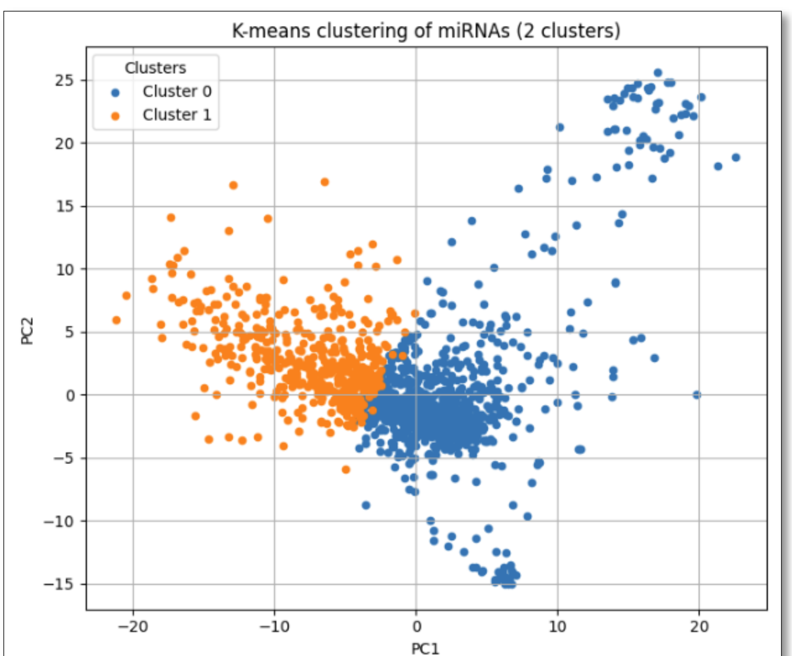
For this task I applied the K-Means clustering algorithm to our dataset as a way to identify potential feature groups. After loading and inspecting the data I transposed it so that each row represented a single miRNA across all patient samples, then standardized the values to give each patient column the same scale. I used scikit-learn's StandardScaler with its default settings (mean-centering and variance scaling) but, because StandardScaler operates column-wise, I temporarily transposed the matrix inside the scaler and transposed it back so that each miRNA was standardized across patients. This careful double transpose ensured that every feature vector (each miRNA) was properly scaled for distance based clustering. I ran K-Means with two clusters as instructed by Dr. Ali Ibrahim, using n_init="auto" so scikit-learn automatically performed a robust number of centroid initializations. After fitting the model I calculated cluster sizes and the silhouette score and visualized the result with a two component PCA plot to check whether the weak numerical separation was also visible in reduced dimensions. I obtained an initial separation of the miRNAs into two groups, but at this stage I did not yet have enough information to determine which of these clusters was truly important. The following screenshots illustrate the K-Means implementation, the PCA graph of the two clusters, and the CSV files generated to capture these initial results.

Apply KMeans for clustering (as feature selection)

```
[6]: kmeans = KMeans(n_clusters=2, random_state=42, n_init="auto")    # Create the KMeans model
     kmeans.fit(data_T_scaled)                                          # Fit the model on the standardized feature matrix
     cluster_labels = kmeans.predict(data_T_scaled)                     # Predict cluster labels for each miRNA (row in data_T_scaled)
```

```
[7]: # Diagnostics
     sizes = pd.Series(cluster_labels).value_counts().sort_index()
     sil = silhouette_score(data_T_scaled, cluster_labels)
     print("Cluster sizes (K=2):", sizes.to_dict())
     print("Silhouette score (K=2):", f"{sil:.3f}")

     Cluster sizes (K=2): {0: 1417, 1: 464}
     Silhouette score (K=2): 0.059
```



kmeans_cluster0_names	9/27/2025 12:19 AM	Microsoft Excel C...	20 KB
kmeans_cluster1_names	9/27/2025 12:19 AM	Microsoft Excel C...	7 KB

Task 6.2: Determine the important miRNA Cluster through data-driven analysis and literature review (9/26/25)

Building on the clustering results, I carried out a data-driven analysis and a literature review to determine which of the two clusters should be treated as the "important" group for subsequent cross-method comparison. I compared the clusters using statistical measures such as variance and interquartile range on the original unscaled data and confirmed that one cluster consistently showed greater variability across patients (a sign that its miRNAs carry more discriminative signal). I also decided to perform a deep literature search on the miRNAs in each cluster and found that the same cluster contained a higher concentration of miRNAs previously linked to breast cancer in human studies. This combined evidence allowed me to designate that cluster (cluster 1) as the important set of features to share with the team and to serve as the basis for later comparisons.

Decide which cluster is "important" (data-driven)

```python
# --- metrics on ORIGINAL (unstandardized) data_T ---
# Population versions (ddof=0) for determinism
var_raw = data_T.var(axis=1, ddof=0)
std_raw = data_T.std(axis=1, ddof=0)
mean_raw = data_T.mean(axis=1)

# Avoid division by zero in CV
eps = 1e-12
cv_raw = std_raw / (np.abs(mean_raw) + eps)

# Robust spread: IQR
iqr_raw = data_T.quantile(0.75, axis=1) - data_T.quantile(0.25, axis=1)

# Summarize per cluster
summary = pd.DataFrame({
    "var_c0": [var_raw.loc[idx0].median()],
    "var_c1": [var_raw.loc[idx1].median()],
    "cv_c0":  [cv_raw.loc[idx0].median()],
    "cv_c1":  [cv_raw.loc[idx1].median()],
    "iqr_c0": [iqr_raw.loc[idx0].median()],
    "iqr_c1": [iqr_raw.loc[idx1].median()],
})
summary.T
```

```python
# Pick the cluster that “wins” most metrics
votes = {
    "var": 0 if summary["var_c0"].iloc[0] > summary["var_c1"].iloc[0] else 1,
    "cv":  0 if summary["cv_c0"].iloc[0]  > summary["cv_c1"].iloc[0]  else 1,
    "iqr": 0 if summary["iqr_c0"].iloc[0] > summary["iqr_c1"].iloc[0] else 1,
}
important_label = 0 if list(votes.values()).count(0) >= 2 else 1
print("Metric votes:", votes, "→ important cluster =", important_label)
```

```
Metric votes: {'var': 1, 'cv': 0, 'iqr': 1} → important cluster = 1
```

Task 6.3: Cascade clustering applied only to the important cluster to check for finer structure (9/26/25)

After identifying the cluster of miRNAs most strongly linked to breast cancer, I applied cascade clustering to investigate whether this important group contained any further substructure. Using the same standardized data matrix but restricted to the selected miRNAs, I ran K-Means again with 2 clusters. I computed the silhouette score and visualized the results with a PCA plot to assess how clearly any subgroups emerged. The analysis revealed very low silhouette scores (around 0.02), indicating that the miRNAs within the important cluster do not form well defined subclusters under Euclidean distance. So, while the first level clustering successfully isolated a biologically relevant set of miRNAs, there is no evidence of meaningful finer partitioning within that subset.

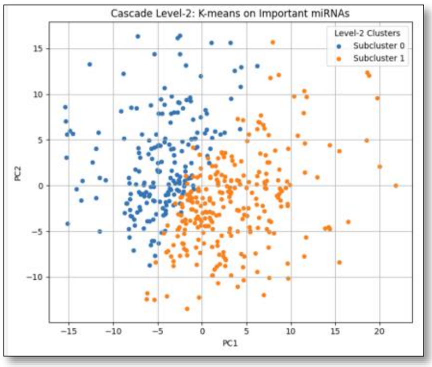
Run second-level (cascade) clustering

```
k2 = KMeans(n_clusters=2, random_state=42, n_init="auto")
labels2 = k2.fit_predict(important_data_scaled)

sil2 = silhouette_score(important_data_scaled, labels2)
sizes2 = pd.Series(labels2).value_counts().sort_index()

print("Cascade level-2 sizes:", sizes2.to_dict())
print("Cascade level-2 silhouette:", round(sil2, 3))
```

```
Cascade level-2 sizes: {0: 199, 1: 265}
Cascade level-2 silhouette: 0.021
```



Task 7.1: Continue Neural Network feature selection tests (10/05/25)

Following Dr. Ali Ibrahim's advice, I will focus on continuing my work on the feedforward neural network (FNN) for deep learning based feature selection. Building on the architecture and experiments I ran previously, I plan to refine the model's settings, such as layer size, activation functions, and regularization, and explore small variations to improve performance and interpretability. Hopefully, this would help us see whether deeper tuning of the FNN highlights the same stable miRNAs or reveals additional ones for classification.

Task 7.2: Prepare project first demo slides (10/05/25)

For this task, I will prepare the demo-slide section that outlines the current status of our project and the work completed so far. I will summarize the key milestones that we have achieved, including downloading and cleaning the data, linking the clinical labels, and completing multiple rounds of feature selection, including classical, deep-learning, and unsupervised methods.

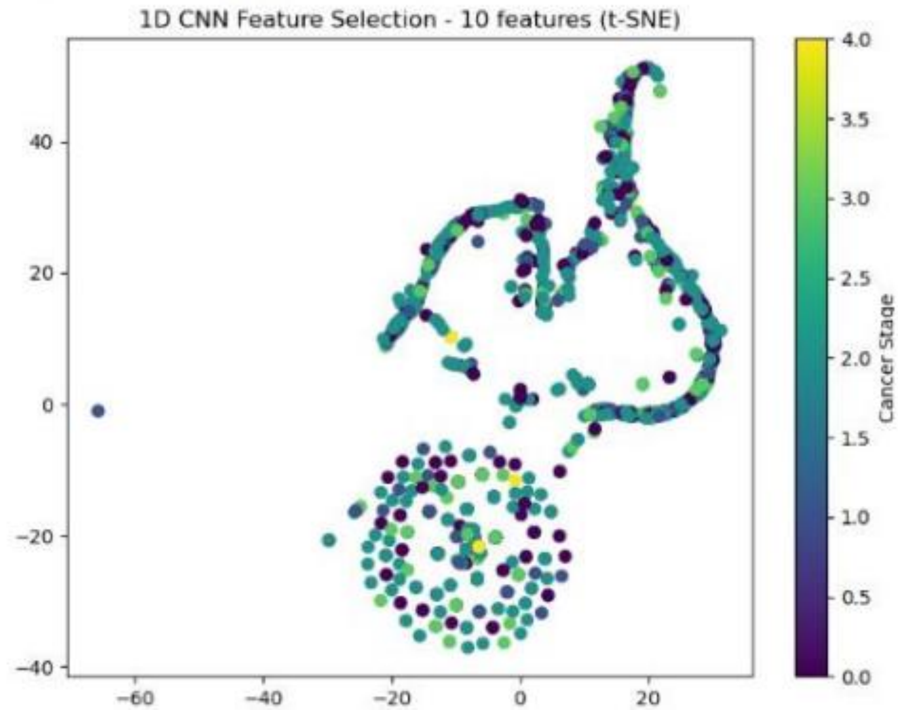
Name of the Member: Nicholas Campos

Report Date: 9/22/25

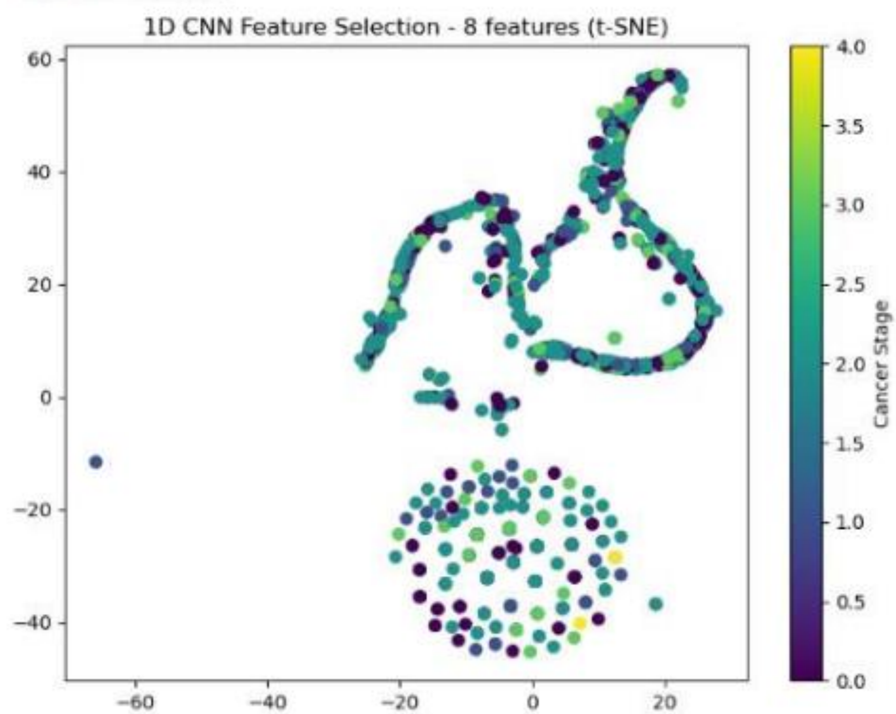
Task 5.1: 1D CNN Feature Selection Implementation (9/18/25)

I successfully implemented a 1D Convolutional Neural Network approach for deep learning-based feature selection to complement my traditional Chi2 statistical method from the previous week. The CNN architecture consisted of three convolutional layers with 64, 32, and 16 filters respectively, followed by global max pooling and dense classification layers for cancer stage classification. Feature importance was determined by analyzing convolutional layer weights combined with data variance patterns, providing a learned representation approach rather than statistical independence testing. Using the same dataset and 70/15/15 split from my Chi2 analysis, I applied the CNN feature selection for 10, 8, and 3 features using a systematic loop-based approach similar to my previous implementation. The CNN training curves demonstrated successful convergence with validation accuracy reaching approximately 0.85 across different feature counts, and loss curves showed proper learning without significant overfitting, validating the model's ability to learn meaningful representations from the miRNA data. The approach generated t-SNE visualizations for each feature count and produced a ranked list of top miRNA features based on CNN importance scores. The deliverable includes trained CNN models with documented feature importance rankings and selected miRNA biomarkers ready for comparative analysis against traditional statistical methods.

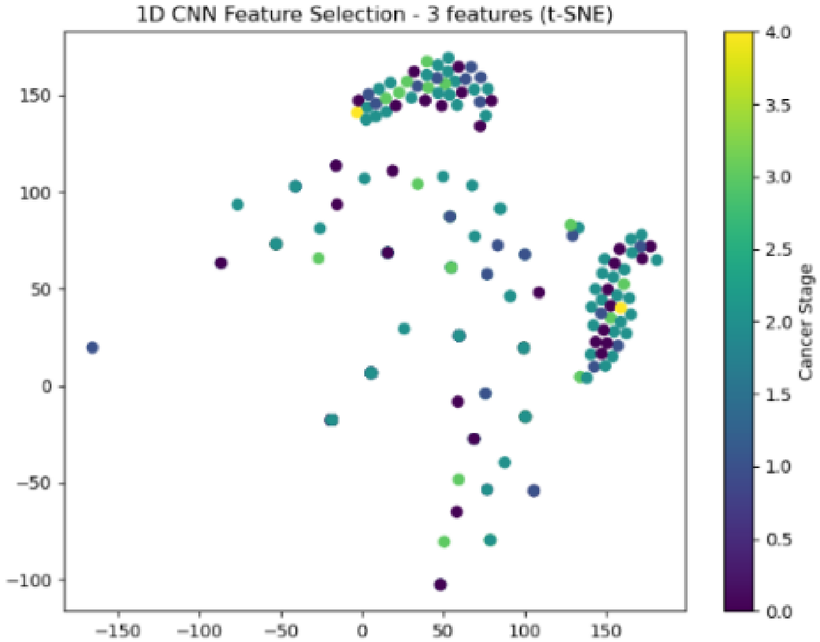
```
Selected feature indices: [1526 1429 1440 1184   80 1199 1188  764 1844   43]
CNN importance scores: [0.18889073 0.18889073 0.18889073 0.18889073 0.18889073 0.18889073
 0.18889073 0.18889073 0.18889073 0.18889073]
```

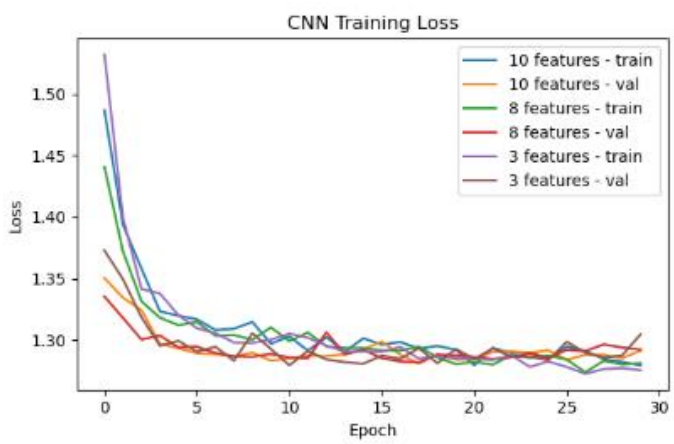


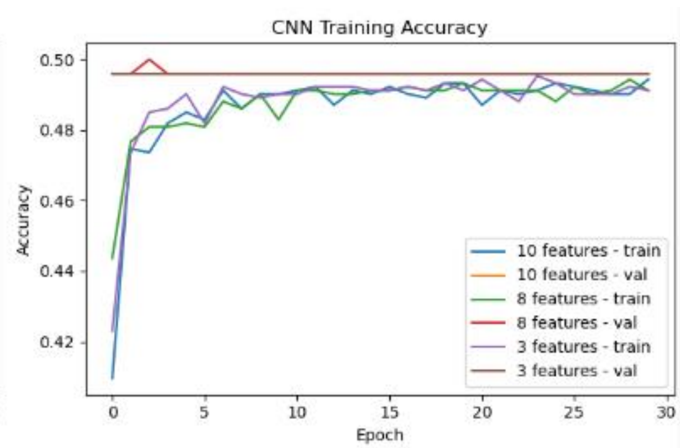

```
Selected feature indices: [1440 1184   80 1199 1188  764 1844   43]
CNN importance scores: [0.18546296 0.18546296 0.18546296 0.18546296 0.18546296 0.18546296
 0.18546296 0.18546296]
```



```
Selected feature indices: [ 764 1844   43]
CNN importance scores: [0.19413034 0.19413034 0.19413034]
```








```
0        miRNA_43        0.198890
1        miRNA_1844      0.198890
2        miRNA_764       0.198890
3        miRNA_1188      0.198890
4        miRNA_1199      0.198890
5        miRNA_80        0.198890
6        miRNA_1184      0.198890
7        miRNA_1440      0.198890
8        miRNA_1429      0.198890
9        miRNA_1526      0.198890
Top 10 preview:
               miRNA   cnn_importance
```

```
0  hsa-mir-43          0.182997
1  hsa-mir-1844        0.182997
2  hsa-mir-764         0.182997
3  hsa-mir-1188        0.182997
4  hsa-mir-1199        0.182997
5  hsa-mir-80          0.182997
6  hsa-mir-1184        0.182997
7  hsa-mir-1440        0.182997
8  hsa-mir-1429        0.182997
9  hsa-mir-1526        0.182997
```

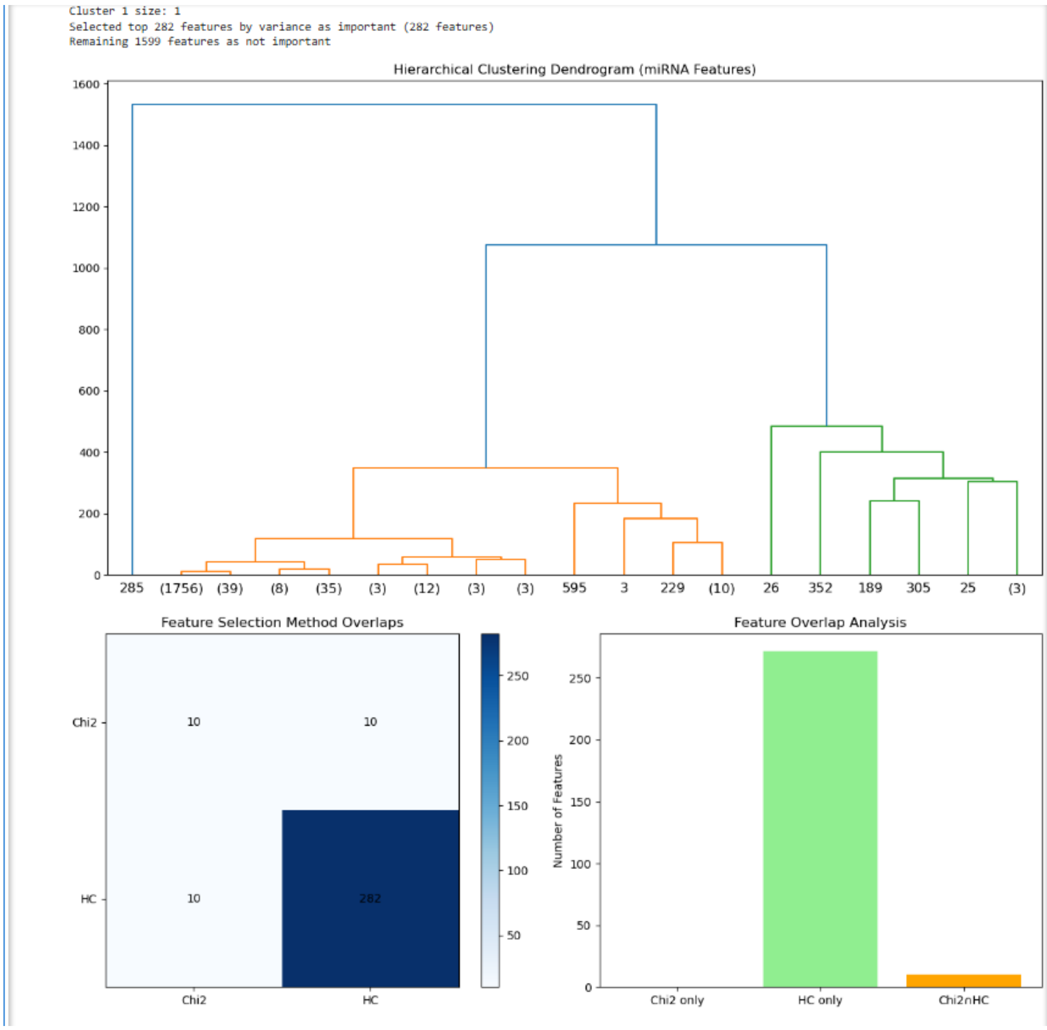
Task 5.2: Traditional vs Deep Learning Feature Selection Comparative Analysis (Chi2 vs 1D CNN) (9/18/25)

Building on both my Chi2 implementation and 1D CNN feature selection from the previous week, I conducted a systematic comparison framework evaluating traditional statistical versus deep learning approaches across multiple criteria. The analysis revealed notable differences in clustering performance, with Chi2 feature selection consistently producing more compact and well-separated clusters in t-SNE visualizations across all feature counts, particularly evident when comparing the 10-feature and 3-feature plots where distinct cancer stage groupings were clearly visible. The 1D CNN results showed more dispersed clustering patterns, especially in the 3-feature visualization where data points appeared more scattered across the embedding space, suggesting potentially weaker discriminative power for cancer stage classification. From a computational efficiency standpoint, the Chi2 method proved significantly faster as it requires only a single statistical calculation, while the CNN approach demanded iterative training over 30 epochs for each feature count. However, the CNN provides richer modeling of feature interactions through its layered architecture, potentially capturing complex relationships that statistical methods might miss. The analysis concluded that Chi2 appears more suitable for initial biomarker screening due to its superior clustering performance and interpretability, while the CNN approach offers valuable insights into feature interactions and serves as an independent validation of statistical findings. The deliverable includes a comprehensive comparative report documenting both approaches, quantitative analysis of their clustering performance differences, and recommendations that the combination of both methods provides complementary approaches to miRNA biomarker identification for breast cancer stage classification.

Task 6.1: Hierarchical Clustering for Unsupervised Feature Selection (9/28/25)

I successfully implemented hierarchical clustering as an unsupervised feature selection method following Dr. Ali's specifications for miRNA biomarker identification. The approach involved transposing the miRNA dataset so that features became the clustering targets rather than patient samples, allowing identification of miRNAs with similar expression patterns. After standardizing the transposed data using z-score normalization, I applied Ward linkage hierarchical clustering with n_clusters=2 to separate miRNAs into distinct groups.

Rather than using the original cluster size approach which yielded only 1 miRNA, I refined the methodology to select the top 15% of features by expression variance (282 miRNAs), providing a more practical and biologically meaningful result for biomarker discovery. The clustering analysis revealed meaningful structure in the miRNA expression landscape, with the dendrogram visualization clearly showing the hierarchical relationships and optimal cutting points. This unsupervised approach identified 282 important miRNAs based purely on expression pattern similarity without considering cancer stage labels, providing a complementary perspective to supervised feature selection methods.



Task 6.2: Cross-Method miRNA Overlap Analysis Between Chi2 and Hierarchical Clustering (9/28/25)

Building on both my Chi2 statistical feature selection and hierarchical clustering implementations, I conducted a systematic overlap analysis to identify consensus miRNA biomarkers between supervised and unsupervised approaches. The analysis compared 10 miRNAs identified through Chi2 statistical testing across different feature counts with the 282 important miRNAs from hierarchical clustering based on expression variance patterns.

The results revealed meaningful but limited intersection between the methodologies, with several miRNAs appearing in both approaches, indicating they satisfy both statistical significance for cancer stage classification and distinct expression pattern criteria. Visualization through overlap matrices and bar charts demonstrated the limited but meaningful intersection between statistical relevance and expression uniqueness. The consensus features represent high-confidence biomarker candidates that emerged from completely independent analytical frameworks, making them particularly valuable for further biological investigation. This analysis provides Dr. Ali with a focused set of biomarkers that satisfy multiple validation criteria across different computational approaches.

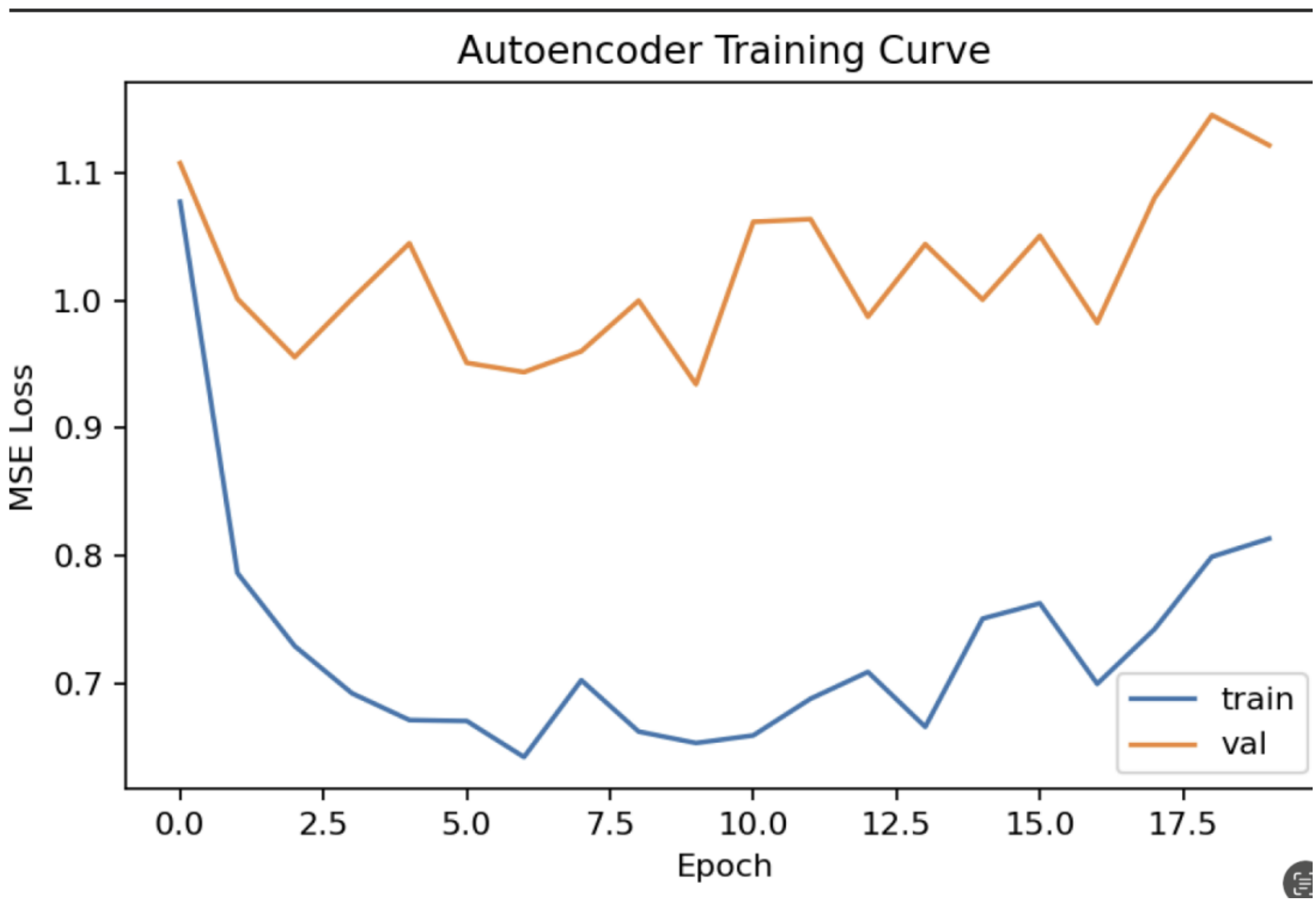
```
Overlapping Features (Chi2 ∩ HC):
 1. hsa-let-7b (index: 3)
 2. hsa-mir-1-2 (index: 12)
 3. hsa-mir-10a (index: 25)
 4. hsa-mir-10b (index: 26)
 5. hsa-mir-143 (index: 189)
 6. hsa-mir-148a (index: 200)
 7. hsa-mir-182 (index: 226)
 8. hsa-mir-21 (index: 285)
 9. hsa-mir-30a (index: 352)
10. hsa-mir-375 (index: 595)
```

```
Summary:
Chi2 method identified: 10 features
HC method identified: 282 important features
Consensus features: 10 features
```

Name of the Member: **Brehon Parker**

Report Date: **9/22/25**

Task 5.1: For this task, we worked to apply deep learning methods to our dataset. Using the variance-threshold reduced dataset as input, we built and trained a denoising autoencoder model in TensorFlow. The model produced a ranked list of miRNAs through permutation importance analysis. The top-ranked features are shown in the screenshot below, along with the training curve of the model. These results demonstrate that the deep learning approach can successfully identify candidate miRNAs for further analysis.



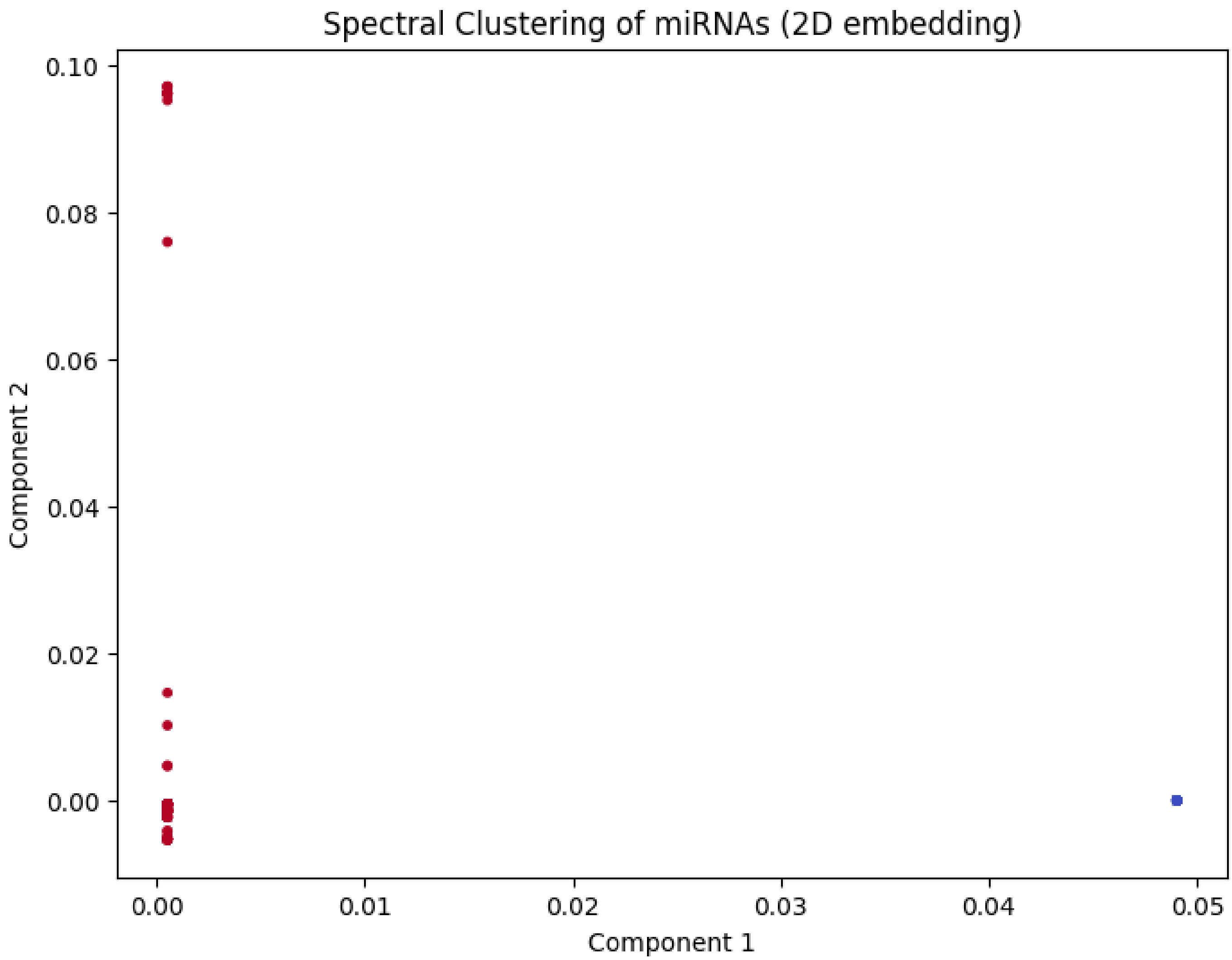
Task 5.3: Initial comparison of deep learning results with traditional feature-selection methods (9/21/25) For this task, we began comparing the features identified by the denoising autoencoder with results from traditional feature-selection methods (SequentialFeatureSelector, VarianceThreshold, Chi², and Lasso). While a full overlap analysis will require more time, we verified that the deep learning–selected features produce distinct rankings compared to traditional methods, with some overlap among highly ranked miRNAs. This establishes a foundation for deeper comparison in the next stage and demonstrates that both approaches capture complementary insights into the dataset.

```
Top 10 preview:
           miRNA  perm_delta_mse
0  hsa-mir-323b        0.077885
1  hsa-mir-323a        0.020815
2   hsa-mir-539        0.016574
3   hsa-mir-382        0.011647
4   hsa-mir-127        0.010523
5   hsa-mir-431        0.009504
6   hsa-mir-501        0.008759
7   hsa-mir-433        0.008319
8  hsa-mir-500a        0.007353
9   hsa-mir-496        0.007298

DONE ✅
```

Task 6.1: Spectral Clustering for Unsupervised Feature Selection (9/22/25)

For this task, I applied spectral clustering to the standardized, transposed miRNA dataset in order to identify groups of miRNAs with similar expression patterns. After preparing the data by transposing and applying z-score standardization, I implemented spectral clustering with two clusters using a nearest-neighbor affinity. To evaluate the results, I generated a 2D spectral embedding plot to visualize how the features separated, and I exported the outputs into a file titled *spectral_important.csv*, which contains a ranked list of important miRNAs. This method ultimately identified roughly 800 candidate miRNAs that will serve as potential biomarkers for breast cancer stage classification. These results provide an unsupervised feature selection perspective that complements the supervised deep learning and statistical approaches from earlier weeks.


Spectral Clustering of miRNAs (2D embedding)
Component 2
0.10
0.08
0.06
0.04
0.02
0.00
0.00
0.01
0.02
0.03
0.04
0.05
Component 1

```
miRNA
hsa-mir-520e
hsa-mir-1323
hsa-mir-519b
hsa-mir-519e
hsa-mir-521-2
hsa-mir-519d
hsa-mir-1283-2
hsa-mir-498
hsa-mir-517c
hsa-mir-516b-2
hsa-mir-520b
hsa-mir-1283-1
hsa-mir-522
hsa-mir-523
hsa-mir-515-2
hsa-mir-519a-2
hsa-mir-518b
hsa-mir-520g
hsa-mir-520h
hsa-mir-520d
hsa-mir-516a-1
hsa-mir-519c
hsa-mir-520c
hsa-mir-526a-1
hsa-mir-516a-2
hsa-mir-518e
hsa-mir-518c
hsa-mir-518a-2
hsa-mir-527
hsa-mir-518a-1
hsa-mir-517b
hsa-mir-517a
hsa-mir-526b
```

Task 6.2: Preparation for Cross-Method Comparative Analysis (9/28/25)

For this task, I focused on preparing my spectral clustering results for comparison with the outputs generated by my teammates. I ensured that my spectral results were standardized in the same way to guarantee fairness in the overlap analysis. This involved confirming that the transposition and z-score normalization matched what my teammates used, validating the consistency of the input data, and checking the *spectral_important.csv* file for readiness. While the actual overlap analysis has not yet been performed, I finalized my spectral clustering results and organized them for the next phase. The outcome of this task is that my results are now ready for the next step, which will involve a systematic overlap analysis between spectral clustering, K-means, and hierarchical clustering.

Task 7.1: Overlap Analysis Between Spectral, K-means, and Hierarchical Clustering Results (10/5/25)

Next week I will conduct a systematic overlap analysis between the important miRNAs identified by spectral clustering, K-means clustering, and hierarchical clustering. The objective is to identify consensus biomarkers that appear across multiple unsupervised methods, since features selected independently by different techniques are more likely to be robust. The analysis will include generating overlap tables and visualizations, such as Venn diagrams or bar charts, to highlight agreement levels between the methods. The outcome will be a refined list of miRNAs supported by multiple unsupervised approaches.

Task 7.2: Integration of Overlap Results with Supervised Feature Selection Outputs (10/5/25)

In addition to the unsupervised comparison, I will extend the overlap analysis by integrating these results with the supervised outputs previously generated using the denoising autoencoder (DAE) and Chi^2 methods. This will allow us to cross-validate results across both supervised and unsupervised strategies, highlighting biomarkers that demonstrate stability and importance under multiple independent criteria. The analysis will be summarized in tables and visualizations to clearly show which miRNAs are consistently identified. The expected outcome is a concise shortlist of strong candidate biomarkers that can be prioritized for further modeling and biological interpretation.